

Interpret Diffusion Based Global Style Transfer

Jeongha Lee

1. About Self-Discovering Interpretable Diffusion Latent Directions for Responsible Text-to-Image Generation :

This paper investigates how text-to-image diffusion models generate biased or unsafe content and proposes a method to directly identify and control the internal semantic representations that cause these issues. While prior work focuses on prompt filtering, safety classifiers, or fine-tuning, these approaches do not explain how the model internally encodes concepts related to gender, race, sexuality, or violence. The authors instead examine the U-Net bottleneck layer—known as the h-space—which functions as a semantic latent space.

The key contribution is a self-supervised method for discovering latent directions that correspond to arbitrary concepts without using labels or external models. To learn a concept such as “female,” the model first generates images from a prompt containing that concept, then reconstructs those images using a modified prompt with the concept removed. Because the diffusion model lacks the missing concept in the new prompt, it must rely on a learnable vector inserted into the h-space. Optimizing this vector forces it to represent the removed semantic information. This mechanism generalizes to many concepts, including fairness-related and safety-related attributes.

The learned vectors enable several forms of responsible generation. For fair generation, sampling gender or race vectors uniformly during inference reduces biased outputs—for example, ensuring that prompts like “doctor” produce balanced male and female images. Experiments on the Winobias benchmark show that this approach outperforms previous debiasing methods and remains robust even when biased wording is added to prompts.

For safe generation, vectors such as “anti-sexual” or “anti-violence” suppress inappropriate content while preserving prompt fidelity. When combined with existing safety systems like SLD or ESD, the method further reduces harmful outputs on the I2P benchmark.

The approach also enhances responsible text guidance: when prompts include phrases like “no violence,” the model often fails to enforce them, but injecting the corresponding safety vector leads to images that better follow such instructions.

Finally, the authors show that these concept vectors exhibit smooth interpolation, can be composed to control multiple attributes simultaneously, and generalize beyond the training context. Importantly, they do not significantly degrade image quality.

Overall, this work demonstrates that diffusion models contain interpretable latent directions for

ethical and semantic concepts, and manipulating these directions provides a lightweight and effective strategy for improving fairness, safety, and responsibility in text-to-image generation.

2. Training Code Analysis

2-1) Create the synthetic dataset with Stable Diffusion

```
23 class DataCreator:
24     def __init__(self, cfg):
25         self.root_dir = cfg.root_dir
26         self.image_prompt = repeat_ntimes(cfg.image_prompt, cfg.num_samples)
27         self.input_prompt_and_target_concept = repeat_ntimes(cfg.input_prompt_and_target_concept, cfg.num_samples)
28         self.validation_prompt_and_concept = cfg.validation_prompt_and_concept
29         print(f"to create {len(self.image_prompt)} total number of samples in {cfg.root_dir}")
30
31     def create_images(self, num_inference_steps=30):
32         pipe = StableDiffusionPipeline.from_pretrained(
33             "stabilityai/stable-diffusion-2-1",
34             )
35         pipe = pipe.to("cuda")
36         pipe.safety_checker=None
37         pipe.set_progress_bar_config(disable=True)
38
39         os.makedirs(self.root_dir, exist_ok=True)
40         for idx, prompt in tqdm(enumerate(self.image_prompt), total=len(self.image_prompt)):
41             if isinstance(prompt, list) or isinstance(prompt, tuple):
42                 output = pipe(prompt[0], negative_prompt=prompt[1], num_inference_steps=num_inference_steps, return_dict=True)
43                 print("positive : ",prompt[0])
44                 print("negative prompt :",prompt[1])
45             else:
46                 print("prompt : ",prompt)
47                 output = pipe(prompt, num_inference_steps=num_inference_steps, return_dict=True)
48                 image = output[0][0]
49                 image.save(self.root_dir+"/"+f"{idx}.jpg")
50
51     def create_labels(self,):
52         os.makedirs(self.root_dir, exist_ok=True)
53         json.dump(self.input_prompt_and_target_concept, open(self.root_dir + "/labels.json", "w"))
54         json.dump(self.validation_prompt_and_concept, open(self.root_dir + "/test.json", "w"))
55         json.dump(update_concept_dict(), open(self.root_dir + "/concept_dict.json", "w"))
56
57     def run(self):
58         self.create_labels()
59         self.create_images()
60
61
62 class Cfg:
63     root_dir="datasets/vangogh_5"
64     num_samples=5
65
66     image_prompt = [
67         "vangogh-styled",
68     ]
69
```

This code implements a synthetic data generation pipeline designed to create an h-space training dataset using Stable Diffusion. The `DataCreator` class initializes configuration parameters, including output directories, prompts, and the number of samples, then procedurally generates images by invoking a Stable Diffusion pipeline for each prompt. It handles both single prompts and paired positive/negative prompt formulations, where the positive prompt represents the desired target concept (e.g., *"vangogh-styled"*) and the negative prompt represents a neutral or concept-free description (e.g., *"A photo"*) to discourage the model from producing the target concept. The system saves the resulting images to the designated dataset directory and records metadata such

as input prompts, target concepts, and validation prompts into JSON label files. Through its `run()` method, the system first writes all label information and then synthesizes the corresponding images, producing a complete synthetic dataset suitable for training the h-space to capture the artist's various artworks.



Figure 1 Synthetic van-gogh styled images via Stable Diffusion with positive prompt

2-2) Train the H-space with synthetic dataset

```

271     print("Start training")
272     loss_history=[]
273     train_loss = 0.0
274     curious_time=0
275     global_step = 0
276     for epoch in range(args.num_train_epochs):
277         unet.train()
278         for step, batch in enumerate(train_dataloader):
279             latents = vae.encode(batch["pixel_values"].to(weight_dtype)).latent_dist.sample()
280             latents = latents * 0.18215
281
282             noise = torch.randn_like(latents)
283             bsz = latents.shape[0]
284             timesteps = torch.randint(0, noise_scheduler.config.num_train_timesteps, (bsz,), device=latents.device)
285             timesteps = timesteps.long()
286
287             noisy_latents = noise_scheduler.add_noise(latents, noise, timesteps)
288             encoder_hidden_states = text_encoder(batch["input_ids"].to(device))[0]
289             if noise_scheduler.config.prediction_type == "epsilon":
290                 target = noise
291             elif noise_scheduler.config.prediction_type == "v_prediction":
292                 target = noise_scheduler.get_velocity(latents, noise, timesteps)
293             else:
294                 raise ValueError(f"Unknown prediction type {noise_scheduler.config.prediction_type}")
295
296             model_pred = unet(noisy_latents, timesteps, encoder_hidden_states, controlnet_cond=batch["input_conditions"].to(device)).sample
297             loss = F.mse_loss(model_pred.float(), target.float(), reduction="mean")
298
299             train_loss += loss.item()
300             curious_time += timesteps.sum().item()
301
302             loss.backward()
303             optimizer.step()
304             lr_scheduler.step()
305             optimizer.zero_grad()
306
307
308             progress_bar.update(1)
309             global_step += 1
310             if global_step%1==0:
311                 train_loss = train_loss/1
312                 accelerator.log({"train_loss": train_loss, "lr": lr_scheduler.get_last_lr()[0]}, step=global_step)
313                 loss_history.append(train_loss)
314                 train_loss = 0.0
315                 curious_time = 0
316
317         logs = {"step_loss": loss.detach().item(), "lr": lr_scheduler.get_last_lr()[0]}

```

This code implements the training procedure for learning the h-space representation (=concept vector) using the synthetic dataset introduced in section 3.1. In each iteration, the model encodes input images into latent variables, perturbs them with Gaussian noise according to the diffusion timestep, and uses the noise scheduler to obtain noisy latents x_t . The model then predicts the

noise residual using the UNet conditioned on both the negative prompt (y^-). (e.g., "A person") and the learnable concept embedding c (e.g., "vangogh-styled"), following the objective:

$$c^* = \underset{c}{\operatorname{argmin}} \sum_{x, y \sim D} \sum_{t \sim [0, T]} \|\epsilon - \epsilon_\theta(x_t^+, t, \pi(y^-), c)\|^2$$

The predicted noise $\epsilon_\theta(x_t^+, t, \pi(y^-), c)$ is compared with the ground-truth noise ϵ through an L2 loss, and gradients are backpropagated to update only the learnable concept vector c while keeping the diffusion model frozen. This process effectively aligns the model's internal conditioning space so that the learned h-space vector (=optimal concept vector c^*) captures the desired global style concept (e.g., "vangogh-styled"), enabling the network to reproduce the target concept consistently across timesteps during generation.

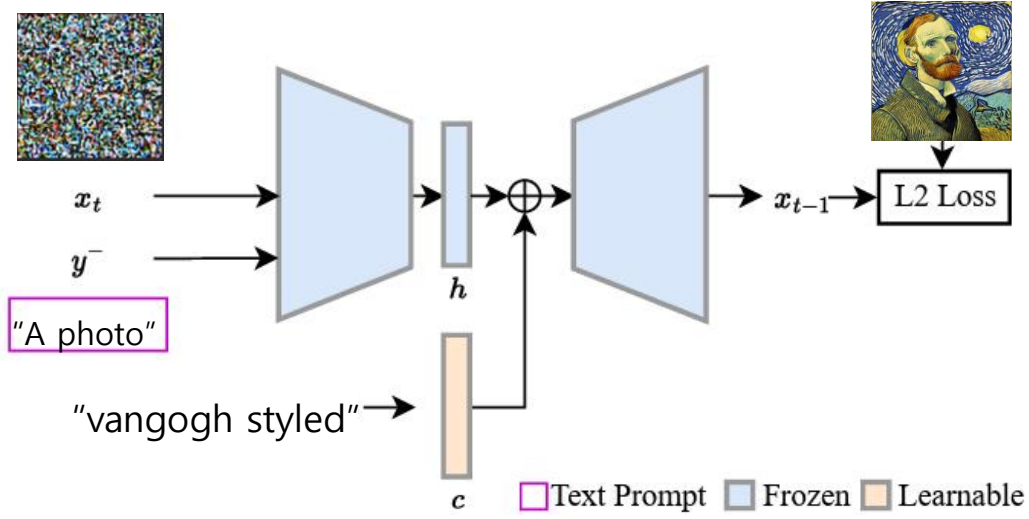


Figure 2 Train the h-space vector (concept vector c) with synthetic dataset

3. Experiment :

This paper presents a new perspective on understanding and manipulating the internal semantic representations of diffusion models, focusing on the bottleneck layer of the U-Net (the h-space), which has been shown to encode structured semantic information. The authors propose a self-supervised method that automatically discovers latent concept vectors—such as gender, race, violence, or sexual content—without relying on external labels or classifiers. By reconstructing images generated with a concept-inclusive prompt using a concept-excluded prompt, the model is forced to store the missing semantic information in an additional learnable vector inserted into the

h-space. These discovered vectors can then be used for fairness, safety, and responsible image generation, offering a lightweight yet effective mechanism for controlling inappropriate or biased content.



Figure 3 The images on the left were generated using the original Stable Diffusion, while the images on the right were generated using Stable Diffusion with the concept vector. The prompt used was “a photo of a building.”

My Global Style Transfer (GST) research builds directly on this structural idea—specifically, the interpretation of the U-Net h-space as a controllable semantic latent representation—and extends it toward the domain of artistic stylization. While InterpretDiffusion focuses on discovering individual concept vectors, my GST framework leverages the same h-space structure but augments it with an MLP-based Style Extraction Layer attached to the U-Net bottleneck. Instead of learning a single latent direction, the MLP is trained on hundreds of artworks from a single artist, allowing it to capture global, multi-exemplar stylistic patterns rather than instance-level attributes. This residual style offset, injected into the h-space at each sampling timestep, encodes high-level artistic characteristics such as color tendencies, brushstroke textures, and compositional deformations unique to the artist.

Whereas InterpretDiffusion discovers explicit semantic concepts (e.g., “female,” “non-violent”) using self-discovery, GST generalizes the same mechanism to the artistic domain by learning a manifold of global style representations across a large corpus of artworks. The h-space’s linearity, semantic consistency, and stability across timesteps—central observations in InterpretDiffusion—serve as foundational design principles for GST. These properties ensure that small residual offsets can steer

the model toward a specific artistic style without compromising content structure or image fidelity.

In essence, InterpretDiffusion demonstrates that ethical and semantic concepts are encoded as interpretable directions in the h-space, while GST extends this insight by showing that complex aesthetic attributes, learned across hundreds of paintings, can also be embedded and controlled within the same latent space. This leads to a novel global style transfer pipeline that moves beyond single-reference style methods and prompt-dependent stylization, enabling stable, consistent, and artist-level global style synthesis directly within the diffusion model's internal representation.

Overall, GST can be viewed as a creative-direction extension of the structural ideas introduced in InterpretDiffusion: both methods manipulate the h-space, but GST transforms this mechanism from a tool for ethical alignment into a framework for rich artistic style modeling. This demonstrates that the h-space is not only an interpretable semantic space but also a powerful unified representation for injecting global stylistic knowledge into diffusion-based image generation.

4. Conclusion :

By extending the structural insights of InterpretDiffusion, our Global Style Transfer (GST) framework demonstrates that the U-Net h-space can serve not only as a semantic latent space but also as an effective representation for global artistic style modeling. Integrating an MLP-based Style Extraction Layer directly on top of the h-space enables the model to learn stylistic manifolds from hundreds of artworks per artist, capturing their global aesthetic tendencies rather than relying on single-reference images or prompt-dependent cues.

Across multiple artists, the learned residual style offsets consistently inject stable and coherent global styles into the generated images without compromising content fidelity. This result aligns with InterpretDiffusion's finding that small, well-structured latent directions can guide semantic outcomes, and further shows that these latent directions can be generalized to high-level, multi-exemplar artistic attributes. Compared to traditional style-transfer approaches, GST produces more consistent stylistic patterns, greater robustness across prompts and seeds, and significantly reduced overfitting to individual artworks.

Overall, these results demonstrate that attaching a lightweight MLP to the diffusion model's h-space enables a scalable and interpretable mechanism for global style learning, effectively expanding InterpretDiffusion's latent manipulation paradigm into the domain of artist-level stylization.